ENGLISH PAPER SUMMARY | ALL ORIGINAL NUMBERED FIGURES

Ocean Wave Interactions with Sea Ice: A Reappraisal

Vernon A. Squire

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Wave-ice models must distinguish conservative scattering from dissipative energy loss, account for the ice regime, and connect wave attenuation to the evolving ice cover. No single attenuation formula resolves every regime.

Prepared independently from the original paper. This is a new English synthesis, not a translation of the Korean notes. The figure atlas reproduces all 5 numbered figures with explanatory reading notes. Section and equation references refer to the source publication.

Research question and scope

This review reassesses the theory, laboratory evidence, field observations, and forecasting applications of ocean wave interactions with sea ice. It concentrates on developments after earlier reviews, especially the Arctic Sea State program. It is a synthesis of published research, rather than a new Great Lakes simulation or a machine learning experiment. Its organizing problem is how to represent the two-way connection between waves and heterogeneous ice at scales usable by forecast and Earth system models. [Sections 1 and 5]

Reduced or less compact ice permits longer fetches and greater wave penetration. Waves can flex and fracture ice, redistribute momentum, and change floe sizes and lateral melting. These processes can modify the ice cover that subsequently attenuates waves. Figure 1 motivates the dynamical relevance of wave stress relative to wind stress near an ice edge.

Physical framework: attenuation is not one process

Scattering redirects wave energy among directions; dissipation removes energy from the resolved wave field and transfers it to other parts of the ice-ocean system. A reduction in a measured directional component cannot, by itself, establish a dissipative loss. Reflection, refraction, directional spreading, ice deformation, overwash, collisions, and turbulence can all matter, with their relative importance changing between ice types. [Section 1.2]

A complex wavenumber encodes dispersion through its real part and exponential amplitude decay through its imaginary part, k_i. Because energy is proportional to squared amplitude, an energy attenuation coefficient is twice the corresponding amplitude attenuation coefficient. This convention must be checked before comparing measurements, parameterizations, or training labels.

Several field datasets exhibit power-law dependence of attenuation on frequency, with exponents commonly around 2-4 (Figure 2). This is an empirical constraint, not a universal fixed exponent. Apparent high-frequency rollover can also reflect wind input and nonlinear spectral transfer, so attenuation inferred without the complete spectral balance can be misleading. [Sections 1.2, 3.3 and 5]

Model families and their assumptions

Continuous and landfast ice is often represented as a thin elastic plate coupled to water motion. Such models resolve flexural-gravity waves, reflection, transmission, and effects of cracks or thickness variations. Their usefulness depends on whether plate assumptions and geometric approximations match the ice being studied. [Section 2.1.1]

For discrete floes, single-floe scattering is a building block for multiple-scattering descriptions. Floe size, spacing, thickness, and wavelength govern the response. Directional transport and strip-based methods offer ways to connect detailed scattering calculations to large-scale models; Figure 3 illustrates incident, reflected, and transmitted wave fields. [Sections 2.1.2-2.1.3]

Frazil, grease ice, and pancakes can be much smaller than a wavelength. In these regimes, viscous or viscoelastic representations may be more appropriate than a scattering-only description. Wang-Shen, Keller, and close-packing models provide different constitutive assumptions. Multiple dispersion roots and non-unique parameter fits complicate interpretation; fitting an observed spectrum does not uniquely identify the physical dissipation mechanism. [Section 2.2; Figure 4]

What the observations establish

Laboratory experiments isolate processes such as plate bending, overwash, and interactions among floes. However, restricted wavelengths, walls, surrogate materials, and mechanical scaling limit direct extrapolation to natural sea ice. Even laboratory-grown ice need not reproduce the material structure of field ice. [Section 3.1]

Field buoys, ship campaigns, satellites, and aircraft provide complementary evidence. Ice and waves vary in both space and time, making a drifting or short-duration record an imperfect sample of the evolving system. Figure 5 compares inferred attenuation spectra and model curves, illustrating both the strong frequency dependence and variability of observed attenuation. [Section 3.2]

Synthetic aperture radar and other remote sensing methods can recover wave information over large areas, but retrieval and inversion assumptions still require in situ validation. A model should therefore be evaluated using several observables, including spectral shape and direction, rather than significant wave height alone. This last evaluation recommendation is the research interpretation of this summary.

Forecasting implications and limitations

The review distinguishes WAVEWATCH III dissipation modules (IC schemes) from scattering modules (IS schemes). A simple ice mask is a very coarse approximation to the physics. More detailed schemes improve the representation of processes, but differ substantially in assumptions and predicted attenuation. The review's module inventory describes the software and research context at publication, not a current software specification. [Section 4 and Table 1]

The main gap is physically credible, computationally practical dissipation parameterization across heterogeneous ice regimes. Better scattering calculations do not automatically solve dissipative losses. Empirical relationships may be useful where mechanistic understanding remains incomplete, provided that calibration conditions and uncertainty are explicit. [Section 5]

Implications for Great Lakes and surrogate research

The review supplies the physical vocabulary for the Great Lakes project, but its polar examples do not validate a freshwater adaptation. Before learning an attenuation or fracture mapping, specify the ice regime, spectral convention, concentration and thickness inputs, and the physical process being approximated.

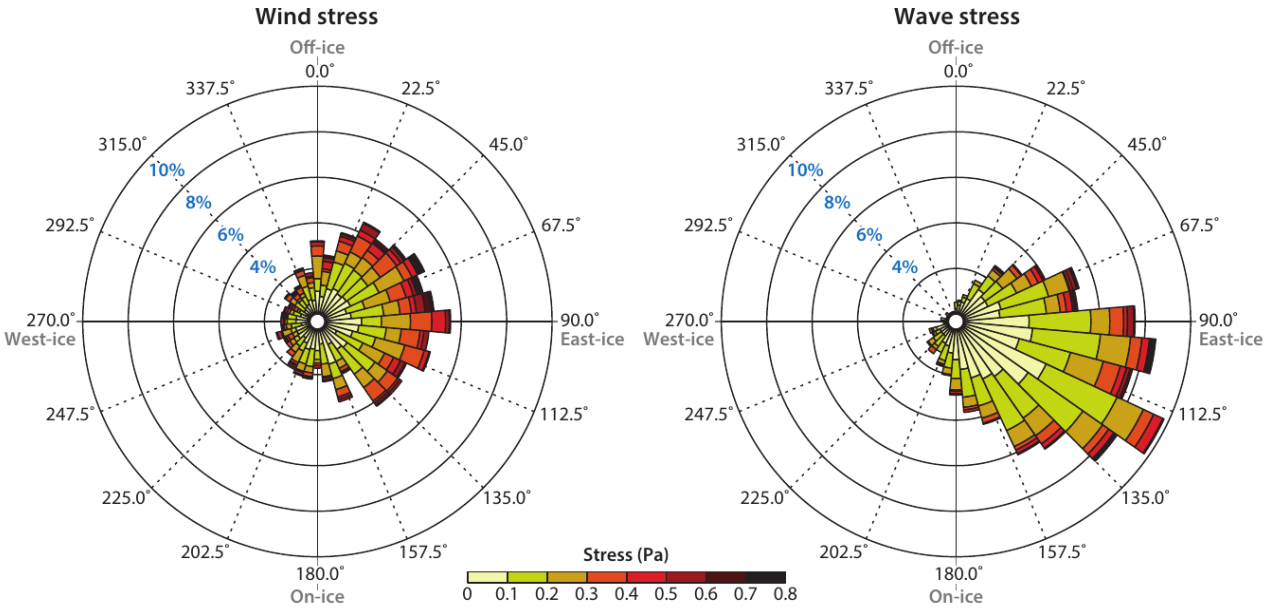
A defensible surrogate should preserve nonnegative energy or probability outputs and be assessed separately for open water, partial ice, and dense ice. Tests should examine changes in spectral shape and physical consistency as well as aggregate error. These are proposed research choices, not experiments performed in the review. The paper supports the need for such a process-based design; it does not itself train an ML surrogate.

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FIGURE ATLAS | 01 / 05

Figure 1. Wind stress and wave stress near the ice edge



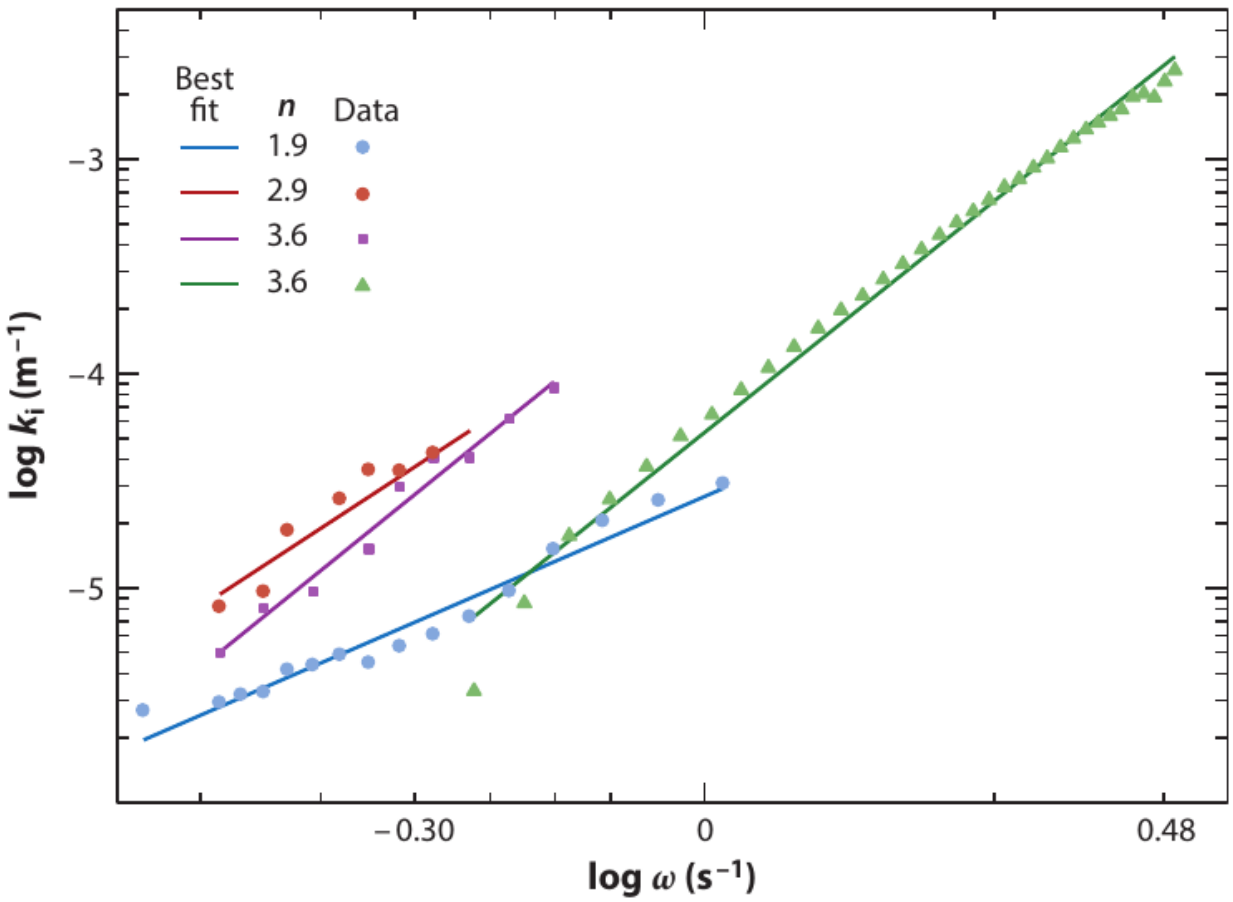
How to read this figure

The two directional distributions compare wind stress from reanalysis with wave stress inferred from Sentinel-1 observations 10 km inside the ice edge. Wave stress has a preferred relationship to the ice edge and can be dynamically important. The figure motivates including wave momentum effects instead of viewing ice only as a passive attenuator. Adapted sources are credited in the original caption: Stopa et al. (2018b).

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FIGURE ATLAS | 02 / 05

Figure 2. Observed frequency scaling of attenuation



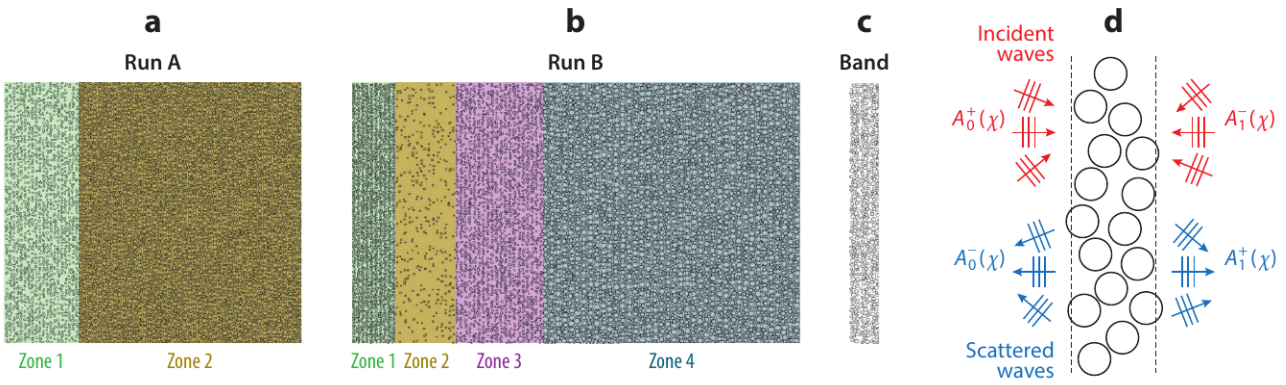
How to read this figure

Four independent field experiments give different fitted slopes for amplitude attenuation versus angular frequency. The figure supports frequency-sensitive parameterizations while showing that a single coefficient or universal slope cannot capture all conditions. It also makes the amplitude-versus-energy attenuation convention important. Original adapted source: Meylan et al. (2018).

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FIGURE ATLAS | 03 / 05

Figure 3. Strip structure and multiple scattering



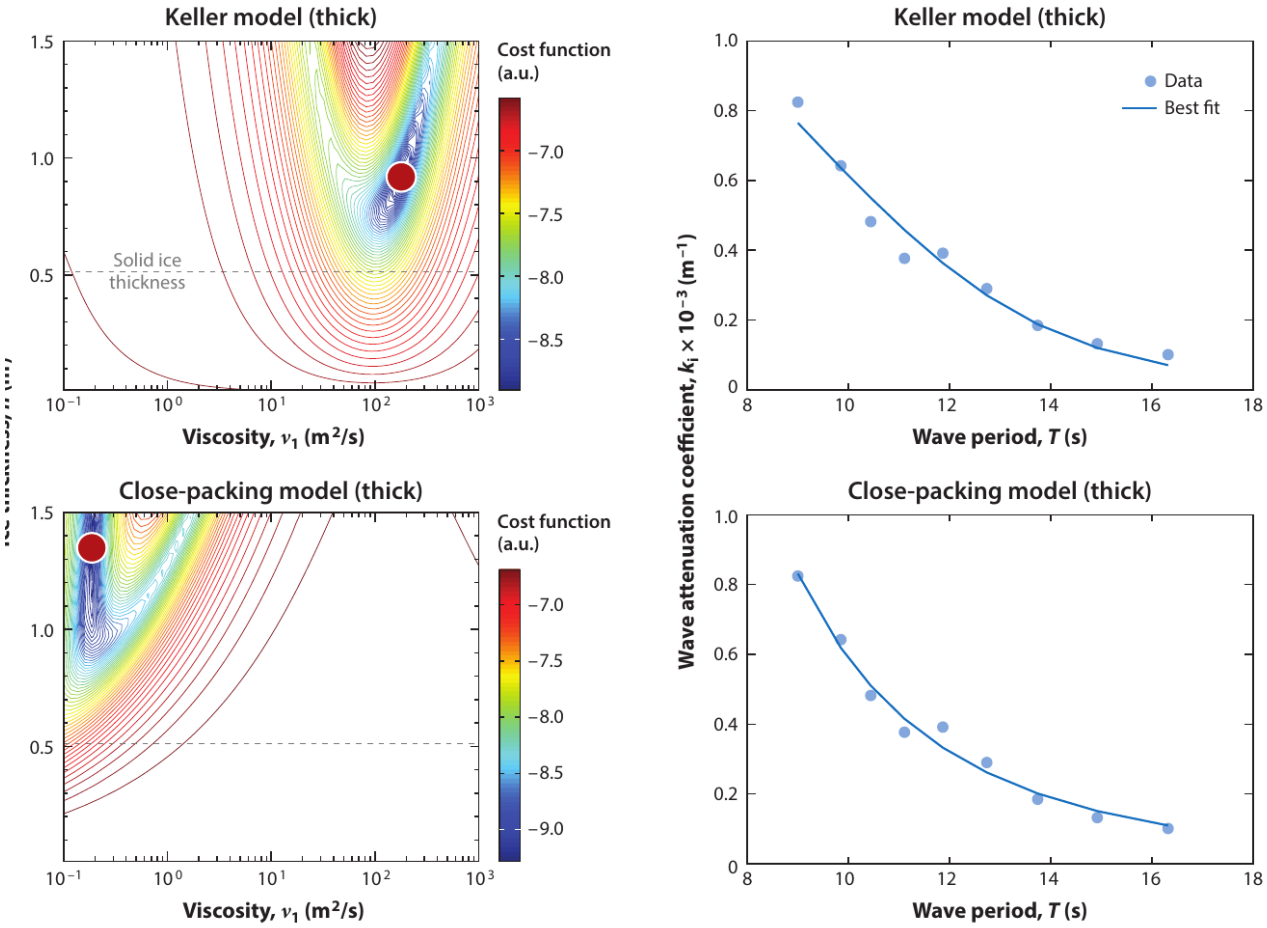
How to read this figure

Panels show examples of marginal ice zone strips and the exchange of incident, reflected, and transmitted directional spectra between neighboring strips. The model preserves the distinction between redistribution and loss. This is a route from single-floe physics toward a tractable representation of a heterogeneous floe field. Original adapted source: Squire and Montiel (2016).

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FIGURE ATLAS | 04 / 05

Figure 4. Fitting viscous and close-packing models



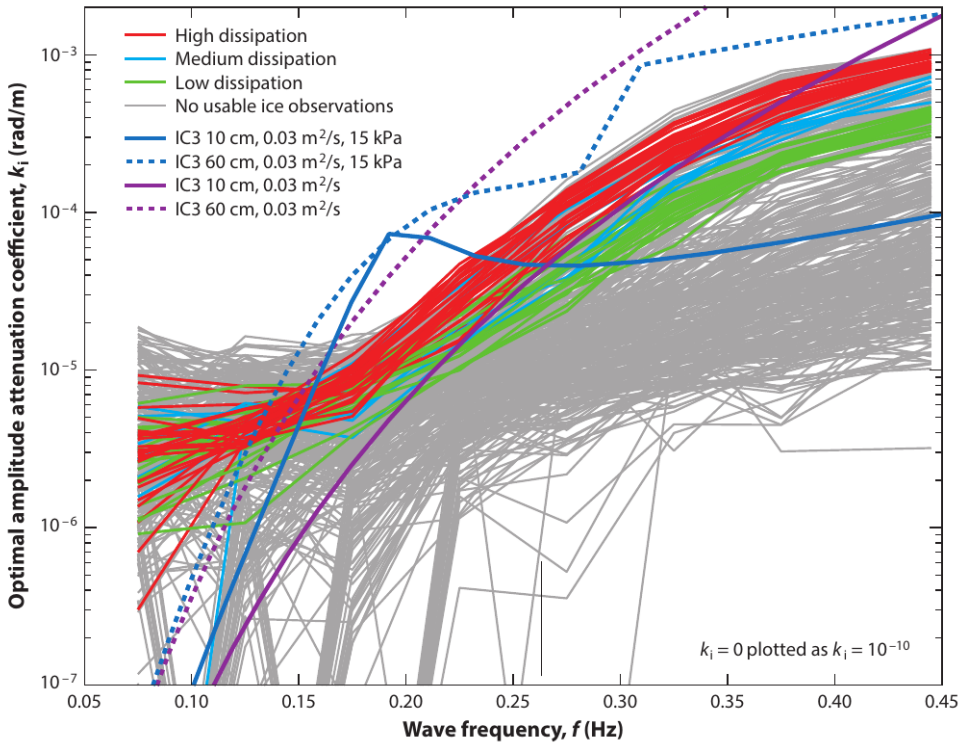
How to read this figure

Cost-function maps and fitted attenuation curves compare Keller and close-packing descriptions against Weddell Sea data. The fitted curves must be read alongside the parameter landscape: an acceptable curve fit need not uniquely constrain viscosity or thickness. The review uses this comparison to discuss the interpretation of effective material parameters. Original adapted source: De Santi et al. (2018).

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FIGURE ATLAS | 05 / 05

Figure 5. Attenuation spectra from the Beaufort Sea



How to read this figure

Observed attenuation curves vary strongly across ice and wave conditions. Viscoelastic-model curves, shown with and without elasticity, provide contrasting spectral dependence. The spread cautions against assuming that a successful fit to one record establishes a general dissipation law. Original adapted source: Rogers et al. (2016).

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